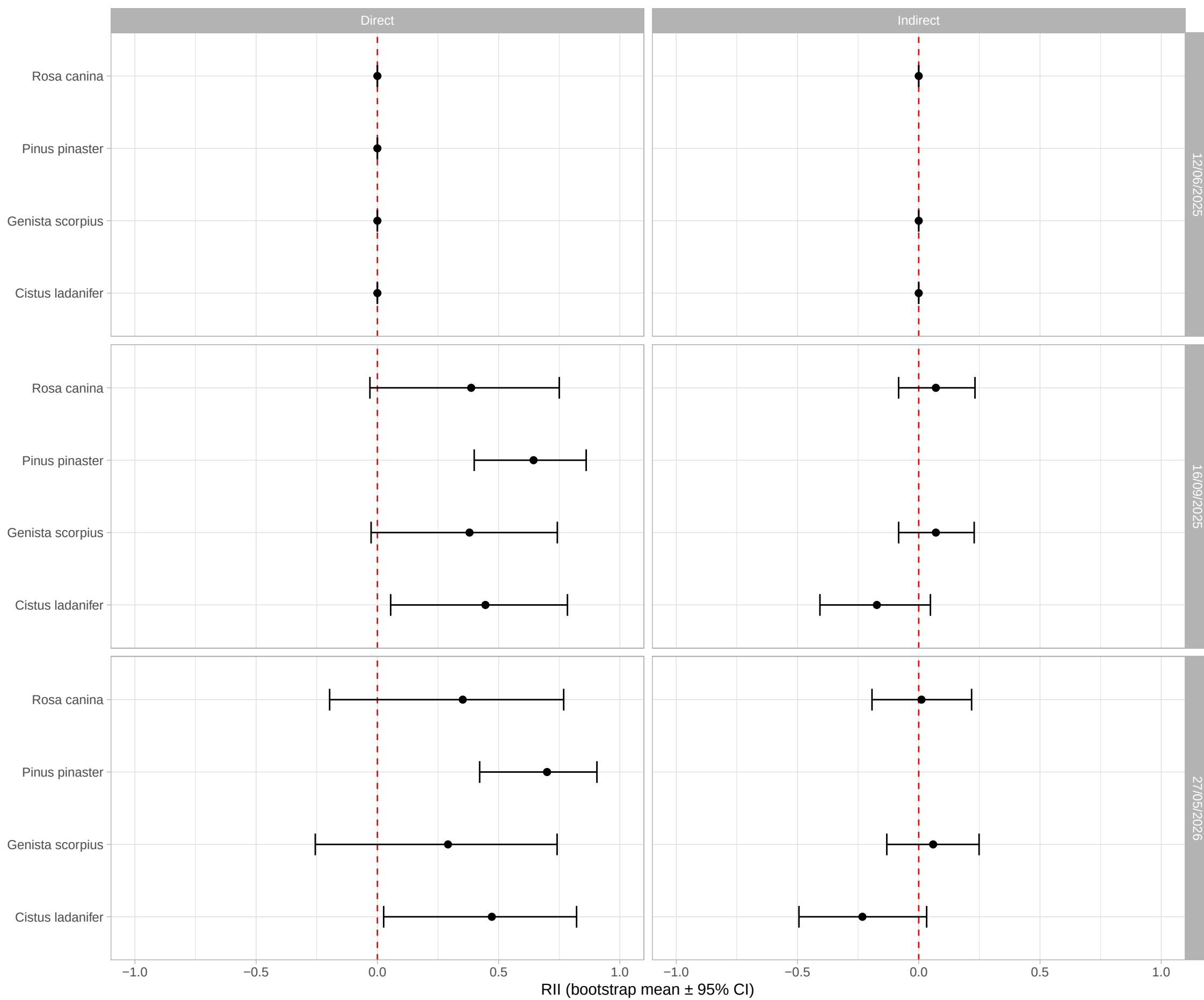
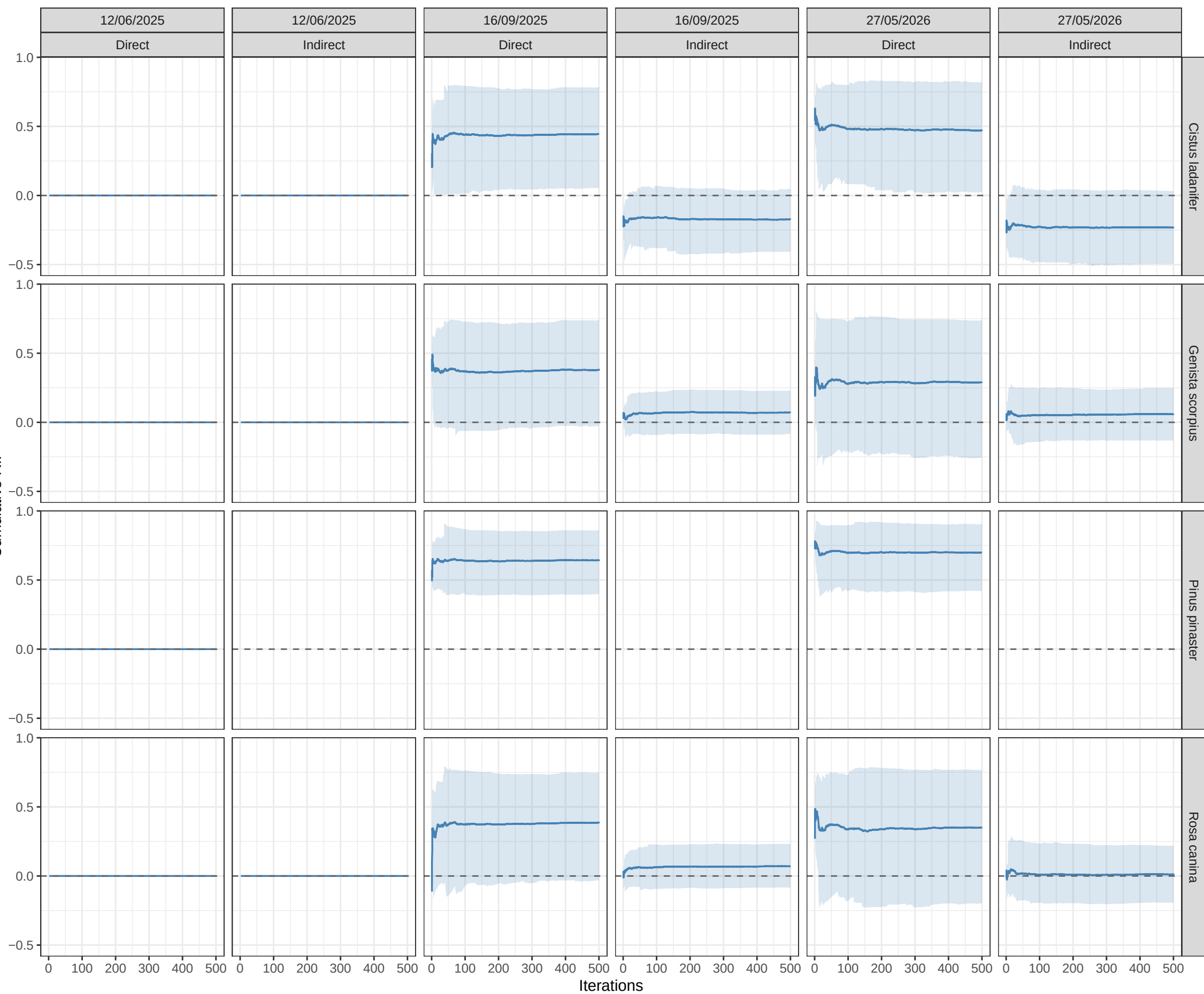


Relative Interaction Index (RII) estimated via GLMM bootstrap



Bootstrap convergence (GLMM-based RII)



## GLMM Bootstrap RII — Methodology and Interpretation

These results were obtained by resampling all individuals (shrubs and controls) with replacement within each combination of environment, microsite, Quercus species and census date. For each of the 500 bootstrap iterations, a GLMM was fitted separately per census date:

$$\text{survival} \sim \text{quercus\_sp} + \text{Environment} * \text{microsite} + (1 \mid \text{Individual})$$

From each fitted model, survival probabilities were predicted for every treatment combination. The RII was then computed from these predicted probabilities for seven biologically meaningful comparisons:

Direct effects of Cistus ladanifer, Rosa canina and Genista scorpius (shrub vs. open in gap). Direct effect of Pinus pinaster (canopy vs. gap in open). Indirect effects of the three shrub species (shrub vs. open under pine canopy).

Plot 1: Bootstrap mean RII and 95% CI for each comparison.  
Values above the dashed red line indicate facilitation;  
values below indicate competition.

Plot 2: Cumulative mean and confidence bounds across iterations, to assess whether 500 iterations are sufficient for stable estimates.

Positive RII = facilitation | Negative RII = competition | RII = 0 = neutral

Bootstrap iterations: 500 | Seed: 123